

Advanced Neuroradiology Center — MRI Suite 3

MRI BRAIN WITH AND WITHOUT GADOLINIUM CONTRAST

SYNTHETIC DOCUMENT — FOR DEMONSTRATION ONLY — CONTAINS NO REAL PATIENT DATA

Patient: Tyler S. Okonkwo

DOB: 05/22/1995

MRN: 6640183

Date: October 29, 2024 at 09:15

Provider: Dr. Marcus Chen, MD — Neuroradiology

CLINICAL INDICATION

Ordering physician: Dr. N. Patel, MD (Neurology). Patient is a 29-year-old male with 6-month history of recurrent severe headaches, predominantly bifrontal, associated with photophobia and mild nausea. No focal neurological deficits on examination. Tension-type headache vs. migraine vs. secondary cause. MRI brain requested to exclude structural etiology.

TECHNIQUE

Field Strength:	3 Tesla (Siemens Prisma)	Sequences:	T1 sagittal, T2 axial, FLAIR axial, DWI, SWI, T1 post-contrast (axial, coronal, sagittal)
Contrast Agent:	Gadavist (gadobutrol) 0.1 mmol/kg IV — no adverse reaction	Study Quality:	Excellent — no motion artifact

FINDINGS

BRAIN PARENCHYMA: No acute intracranial abnormality. No infarct on DWI/ADC sequences. Brain parenchymal signal intensity normal on T1, T2, and FLAIR sequences. No areas of abnormal T2/FLAIR signal hyperintensity in the white matter to suggest demyelination, vasculopathy, or prior ischemic change. Gray-white matter differentiation preserved throughout.

MASS EFFECT AND MIDLINE: No mass, mass effect, or midline shift. Sulci and gyri normal in appearance and distribution for patient age. No herniation.

VENTRICLES AND CSF SPACES: Ventricles normal in size and configuration. No hydrocephalus. Basal cisterns patent. No extra-axial collection.

POSTERIOR FOSSA: Cerebellar hemispheres, vermis, and brainstem morphologically normal. Cranial nerve VII/VIII complex normal bilaterally on thin-slice sequences. No cerebellopontine angle lesion.

VASCULAR: No cerebral cavernous malformation on SWI. No arteriovenous malformation. Caliber of intracranial vessels normal on incidental assessment.

GADOLINIUM POST-CONTRAST: No abnormal parenchymal, meningeal, or ependymal enhancement. No mass with enhancement.

ORBITS AND PARANASAL SINUSES (incidental): Mild mucosal thickening bilateral maxillary sinuses — nonspecific, likely chronic sinusitis. No orbital lesion.

IMPRESSION

1. **NORMAL MRI BRAIN** with and without gadolinium contrast. No structural explanation identified for patient's headache syndrome.

2. Mild bilateral maxillary sinus mucosal thickening — incidental, clinically correlate.

No further neuroimaging indicated at this time. Clinical management should proceed per neurological assessment. Consider headache diary and outpatient neurology follow-up for management of primary headache disorder.

Report dictated and finalized by Dr. Marcus Chen, MD, Certificate of Added Qualification in Neuroradiology, October 29,

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